Decisions Log

This document records key decisions made during the EMS Readiness Optimization project.

Decision Template

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### [DEC-XXX] Decision Title

**Date**: YYYY-MM-DD
**Decision Maker**: [Name/Role]
**Status**: Proposed / Accepted / Superseded

**Context**:
[What situation prompted this decision?]

**Options Considered**:
1. Option A: [Description]
2. Option B: [Description]
3. Option C: [Description]

**Decision**:
[Which option was chosen and why]

**Consequences**:
[What are the implications of this decision?]
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Decisions

[DEC-001] Study Area Selection

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select primary and secondary study areas for the EMS optimization analysis.

Options Considered:
1. All of NYC: Comprehensive but computationally expensive
2. Manhattan only: Focused, manageable scope
3. Manhattan + CBD robustness check: Balanced approach

Decision:
Option 3 - Manhattan as primary study area with CBD (MTA Congestion Relief Zone) as secondary area for robustness testing.

Consequences:
- Allows focused analysis with reasonable computational requirements
- CBD provides natural robustness check for high-demand area
- Results may not generalize to outer boroughs

[DEC-002] Staging Location Candidates

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to define candidate locations for EMS staging.

Options Considered:
1. Grid-based locations: Uniform coverage
2. Hospital locations: Practical operational sites
3. FDNY firehouses: Existing infrastructure
4. Combined approach: Multiple location types

Decision:
Option 3 - Use FDNY firehouses as staging candidates.

Consequences:
- Leverages existing infrastructure
- Realistic operational scenario
- Limited to current firehouse locations (may not be optimal in absolute terms)

[DEC-003] Simulation Framework

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select simulation methodology for EMS system modeling.

Options Considered:
1. Agent-based simulation: Detailed but complex
2. Discrete-event simulation (DES): Standard for queueing systems
3. System dynamics: Too aggregated for this application

Decision:
Option 2 - Discrete-event simulation using SimPy.

Consequences:
- Well-suited for modeling arrival, dispatch, service processes
- Efficient computation
- Python ecosystem enables integration with optimization

[DEC-004] Optimization Approach

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to select optimization method for ambulance staging.

Options Considered:
1. Heuristic approaches: Fast but no optimality guarantee
2. Mixed-integer programming: Optimal but potentially slow
3. Linear programming (relaxed): Fast, near-optimal for many problems
4. Metaheuristics: Flexible but tuning required

Decision:
Option 3 - Linear programming using PuLP, with integer constraints if needed.

Consequences:
- Fast solution times
- Optimality guarantees (for LP formulation)
- May need to round/adjust for integer ambulance counts

[DEC-005] Data Temporal Scope

Date: 2026-03-12
Decision Maker: Project Lead
Status: Accepted

Context:
Need to determine which years of crash data to use.

Options Considered:
1. Most recent year only: Current patterns
2. Last 3 years: Balances recency and stability
3. All available data: Maximum sample size

Decision:
Option 3 - Use all available data for demand pattern estimation, with options to filter by date range.

Consequences:
- Maximum statistical power for pattern estimation
- May include outdated patterns (e.g., pre/post COVID)
- Can subset for sensitivity analysis

Last Updated: March 12, 2026

[DEC-003] Optimization Policy Selection

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Multiple optimization formulations exist for EMS allocation. Need to determine which models to implement and compare for Manhattan EMS readiness.

Options Considered:
1. Demand-Weighted Optimization: Minimize demand-weighted average response time
2. P-Median: Minimize total response time (unweighted)
3. Maximal Coverage: Maximize demand covered within threshold
4. Set Covering: Cover all demand with minimum units
5. Hybrid: Multi-objective optimization

Decision:
Implement and compare options 1-3 (Demand-Weighted, P-Median, Maximal Coverage) plus two baseline policies (Uniform, Demand-Proportional).

Rationale:
- Demand-Weighted (P2) aligns with equity goals (weight by demand intensity)
- P-Median (P2b) provides comparison for unweighted optimization
- Maximal Coverage (P2c) represents pure coverage objective
- Baselines (P0, P1) provide non-optimized benchmarks
- Set Covering is subsumed by other models
- Multi-objective deferred to future work

Consequences:
- Need to implement 3 MIP solvers + 2 baseline policies
- Compare 5 policies × 4 K values = 20 scenarios
- Provides comprehensive policy comparison for decision-makers

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[DEC-004] Unit Count Scenarios (K)

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Need to determine which unit counts to evaluate for optimization and sensitivity analysis.

Options Considered:
1. K = {20, 30, 40, 48} - Current recommendation
2. K = {10, 20, 30, 40, 50} - Wider range
3. K = {25, 30, 35, 40, 45, 50} - Finer granularity
4. K = 40 only - Current Manhattan capacity

Decision:
Use K = {20, 30, 40, 48}

Rationale:
- K=20: Resource-constrained scenario (50% of current)
- K=30: Moderate resource scenario (75% of current)
- K=40: Current Manhattan capacity (baseline)
- K=48: All firehouses staffed with 1 unit (upper bound given capacity=5)
- Provides sufficient range to observe diminishing returns
- Computationally efficient (4 scenarios × 5 policies = 20 optimizations)

Consequences:
- May miss optimal K if it falls outside range (e.g., K=25)

- But results show K=20 achieves near-optimal for P2, so range is sufficient
- Future work can refine with continuous K if needed

[DEC-005] Firehouse Capacity Constraint

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Need to determine maximum units that can be allocated to a single firehouse.

Options Considered:
1. No limit (unconstrained)
2. Capacity = 3 units per firehouse
3. Capacity = 5 units per firehouse
4. Capacity = 10 units per firehouse
5. Variable capacity by firehouse size

Decision:
Use capacity = 5 units per firehouse (uniform across all locations)

Rationale:
- Based on FDNY operational guidance (typical firehouse can support 4-6 units)
- Prevents unrealistic concentration (e.g., all 40 units at one location)
- 5 units allows sufficient flexibility for optimized allocation
- Uniform capacity simplifies optimization formulation
- Conservative estimate (actual capacity varies 3-10 by location)

Consequences:
- P2 allocates up to 5 units at high-demand firehouses (Midtown, Financial District)
- May under-utilize capacity at larger firehouses
- Future work can incorporate actual capacity data per location

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[DEC-006] Coverage Threshold for Maximal Coverage Model

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Accepted

Context:
Maximal Coverage model requires threshold τ (minutes) to define “covered” demand.

Options Considered:
1. τ = 5 minutes (aggressive response target)
2. τ = 8 minutes (NFPA recommendation)
3. τ = 10 minutes (relaxed target)
4. τ = 6 minutes (NYC local law)
5. Multiple thresholds for sensitivity analysis

Decision:
Use $\tau = 8$ minutes (NFPA standard)

Rationale:
- NFPA 1710 specifies 8 minutes for 90% of emergencies
- Aligns with national standards for comparability
- NYC local law (6 min) is aspirational but not consistently achievable
- All policies achieve 100% coverage at $\tau=8$ with $K \geq 20$
- Using $\tau=5$ would require more units; $\tau=10$ is too lenient

Consequences:
- P2c (Maximal Coverage) achieves 100% coverage but with worse response time (3.48 min)
- May want to test $\tau=6$ in future work to align with NYC requirements
- 8-minute threshold appropriate for Manhattan context (dense, short distances)

[DEC-007] Recommended Policy for Deployment

Date: 2026-03-12
Decision Maker: Technical Lead
Status: Proposed

Context:
After comparing 5 policies, need to recommend best allocation strategy for Manhattan EMS.

Options Considered:
1. P0 (Uniform): Equal distribution
2. P1 (Demand-Proportional): Simple heuristic
3. P2 (Demand-Weighted): Minimizes weighted response time
4. P2b (P-Median): Minimizes total response time
5. P2c (Maximal Coverage): Maximizes coverage

Decision:
Primary recommendation: P2 (Demand-Weighted) with K=20-30

Rationale:
- Best response time: 2.49-2.54 min (0-2% from optimal)
- 100% coverage at $K \geq 20$
- Efficient: Uses only 20-23 firehouses vs. 34+ for other policies
- Significant improvement over P0: 17-86% faster response time
- Fast solve time: <0.05 seconds
- Diminishing returns beyond K=30 (additional units provide <0.01 min benefit)

Alternative: P2b for political feasibility (distributes units more evenly)

Consequences:
- Requires MIP solver integration (PuLP + CBC)
- Concentrated allocation may face political resistance (some firehouses get 0 units)
- Recommend phased implementation: Start with P1, transition to P2
- Quarterly reallocation based on updated demand patterns